QCM – Stochastic Models and Simulation

M2 Data Science

Instructions

- Several answers may be correct for each question.
- Tick **all** correct answers; a partially correct selection (missing a correct option or ticking a wrong one) is marked incorrect.
- Duration: **20 minutes**.

1. **Poisson process and thinning.** For a homogeneous Poisson process with rate λ on $[0, T]$:
   (a) The number of arrivals $N(T)$ is Poisson with mean λT.
   (b) Conditionally on $N(T) = n$, the n arrival times are i.i.d. Unif$(0, T)$.
   (c) Thinning with independent Bernoulli(p) marks produces a Poisson process of rate λp.
   (d) Superposition of independent Poisson processes with rates $\lambda_1, \lambda_2, \ldots$ is not Poisson in general.
   (e) Inter-arrival times are i.i.d. exponential with mean λ.
2. **Little's Law and stability.** Consider a single-station queue observed in steady state.
   (a) Little's Law $L = \lambda W$ holds only for $M/M/1$ queues.
   (b) In any stable system, the long-run throughput always equals the *external* arrival rate, even when there is blocking or abandonment.
   (c) If $\lambda \geq c\mu$ in an $M/M/c$, the system is unstable (no finite steady-state mean).
   (d) $L_q = \lambda W_q$ is valid even when service times are non-exponential, provided the system is stable.
   (e) Little's Law cannot be used to validate or cross-check simulation estimates because it is purely theoretical.
3. $M/M/1$ **intuitions and sensitivity.** For an $M/M/1$ queue with arrival rate λ and service rate μ:
   (a) $\rho = \lambda/\mu$ is the utilisation and must satisfy $\rho < 1$ for a stable steady state.
   (b) As $\rho \to 1$, the mean waiting time W_q grows approximately like $1/(1-\rho)$.
   (c) Small relative errors in λ or μ near $\rho \approx 1$ do not significantly affect W_q.
   (d) When ρ is small (e.g. $\rho = 0.2$), W_q is highly sensitive to small changes in λ.
   (e) The stationary distribution of N (number in system) is Poisson with mean ρ.
4. $M/M/c$ **and Erlang-C.** For an $M/M/c$ queue (Poisson arrivals, c exponential servers, infinite buffer):

(a) The offered load is defined as $a := \lambda/(c\mu)$, which equals the utilisation ρ.

(b) The Erlang-C formula gives the probability that an arrival has to wait.

(c) If ρ is fixed and $c \to \infty$, the waiting probability always goes to 1.

(d) For fixed λ, μ, adding servers (increasing c) always decreases W_q with roughly linear (non-diminishing) improvement.

(e) The birth–death structure uses $\lambda_n = \lambda$ and $\mu_n = \min(n, c)\mu$.

5. **CLT-based confidence intervals for simulation output.** You simulate R independent replications of a performance metric X (e.g. mean waiting time) and obtain sample mean $\bar{X}$ and sample standard deviation s.

   (a) A 95% CI can be approximated by $\bar{X} \pm 1.96\, s/\sqrt{R}$ when R is not too small.

   (b) Increasing R shrinks the CI width roughly like $1/\sqrt{R}$.

   (c) Autocorrelation within each replication makes it impossible to apply the CLT and construct meaningful CIs.

   (d) If you use common random numbers across configurations, you cannot construct valid CIs.

   (e) When the distribution of X is heavy-tailed, CLT-based CIs converge just as fast as in the Gaussian case and require no extra caution.

6. **Variance reduction and experiment design.** Consider Monte Carlo or queueing simulation experiments.

   (a) Antithetic variates use positively correlated samples (e.g. U and U') to reduce variance.

   (b) Control variates exploit a correlated quantity with known expectation to reduce estimator variance.

   (c) Using the same random number seeds for two competing system designs can reduce the variance of their *difference*.

   (d) Discarding a warm-up period has no real impact on bias in steady-state estimates.

   (e) Stratified sampling is useless in simulation and should be avoided.

7. **Parameter estimation from event data.** You observe events from a Poisson process and an $M/M/1$ queue.

   (a) For a homogeneous Poisson process observed on $[0, T]$, the MLE for λ is $\hat{\lambda} = N(T)/T$.

   (b) In an $M/M/1$ with observed busy time $B(T)$ and completions $C(T)$, $\hat{\mu} = B(T)/C(T)$ is a natural estimator of μ.

   (c) If you only observe the aggregate count $N(T)$, you cannot consistently estimate λ.

   (d) For $M/G/1$, estimating λ and the first two moments of service times is enough to plug into the Pollaczek–Khinchine formula.

   (e) Fisher information is only relevant for Bayesian estimators and does not help quantify precision for MLEs.

8. **Diagnosing and fitting from an $M/G/c$ log.** You are given an event log with `arrival_time`, `start_service_time`, `completion_time`, `service_time`, `wait_time`, `system_time`, and `queue_len_at_arrival` for a multi-server system.

   (a) By examining overlaps between `start_service_time` and `completion_time`, you can infer the maximum observed number of simultaneous services and thus a lower bound on c.

(b) A histogram of inter-arrival times plus overdispersion of counts can reveal deviations from a homogeneous Poisson arrival process.

(c) A log-survival plot of `service_time` can indicate a shifted exponential structure (flat region then exponential tail).

(d) If you see many arrivals with `queue_len_at_arrival = 0`, the system must have only one server.

(e) If the fitted model consistently underestimates empirical waiting times, it means the model is overfitting noise and is therefore preferable.

9. **Model selection and structural assumptions.** You have a system where customers arrive roughly at rate λ and are processed by several servers; some customers abandon the queue when waiting too long. You fit a simple $G/G/c$ model *without* abandonment based only on completed jobs.

   (a) Using completions only to estimate λ will generally underestimate the true arrival rate.

   (b) Ignoring abandonment tends to make the system appear *more* congested than it really is for arrivals.

   (c) A no-abandonment model fitted on completions may still match the observed waiting times of served customers while completely missing the cost of lost customers.

   (d) If the fitted no-abandonment model reproduces the empirical mean wait of served customers, then it is safe to use it for staffing decisions.

   (e) A key diagnostic is to look for timeouts/reneging patterns (e.g. maximum observed waits, censored records) and decide whether a no-abandonment model is structurally acceptable.

10. **Designing robust simulation studies.** You need to compare two staffing policies for a queueing system using simulation, and report a recommendation with uncertainty.

    (a) Using different warm-up lengths and run lengths for each policy makes comparisons fairer.

    (b) Common random numbers across policies reduce variance of the *difference* in performance, making it easier to detect meaningful improvements.

    (c) When utilisation is very high (ρ close to 1), you should run shorter simulations because queues explode and CLT approximations become unreliable.

    (d) A good practice is to report both point estimates and CIs for key metrics, and to highlight when CIs for competing policies overlap strongly.

    (e) If a small change in the fitted service-time distribution (e.g. slightly heavier tail) flips the ranking of policies, this indicates the decision is *not* robust and should be reported as such.

Correction (do not distribute during exam)

- Q1: a, b, c
- Q2: c, d
- Q3: a, b
- Q4: b, e
- Q5: a, b
- Q6: b, c
- Q7: a, d
- Q8: a, b, c
- Q9: a, c, e
- Q10: b, d, e

Scoring rule (out of 5 points)

- For each of the 50 statements (10 questions $\times$ 5 options):
  - If the statement is *true* and the student ticks it: $+1$ point.
  - If the statement is *true* and the student does *not* tick it: -1 point.
  - If the statement is *false* and the student does *not* tick it: $+1$ point.
  - If the statement is *false* and the student ticks it: -1 point.
- Let S be the total over the 50 decisions ($S \in [-50, 50]$). The QCM mark on 5 points is

$$\text{QCM_score} \;=\; \max\bigl(0,\; \min\bigl(5,\; S/10\bigr)\bigr).$$

  Thus a random paper has expected score close to 0, and only clearly better-than-random performance yields positive points.